

CMP-01 - Constitutional Kernel

Admit or reject protected actions and protected state transitions while preserving HAL's constitutional identity.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 01, 03, 05, 20, 29, 35. Authoritative state: Constitutional Identity Record, Constitution and Invariant Registry, protected-policy admission metadata, Owner Authorization Ceremony Registry, Protected Audit Anchor, Recovery Admission Record. Explicitly non-owned: natural-language interpretation, general planning, ordinary policy administration, general evidence custody.

Critical invariant

No protected effect occurs without current identity, authority, policy, evidence, verification, and exact-change checks.

Logical interfaces

CMP-01-IF-01 / Command / Evaluate Protected Action

Command `Evaluate Protected Action` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-02 / Command / Commit Protected Mutation

Command `Commit Protected Mutation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-03 / Command / Validate Owner Authorization

Command `Validate Owner Authorization` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-04 / Command / Admit Constitutional Recovery

Command `Admit Constitutional Recovery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-05 / Query / Get Constitutional State

Query `Get Constitutional State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-06 / Query / Get Protected Decision

Query `Get Protected Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-07 / Event / Protected Action Admitted

Event `Protected Action Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-08 / Event / Protected Action Denied

Event `Protected Action Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-01-IF-09 / Event / Constitutional Mode Changed

Event `Constitutional Mode Changed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Normal > Restricted > Safe > Safe Recovery > Normal. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: root-state uncertainty, Owner identity conflict, replayed Authority or Permission decision, audit-anchor mismatch, replica disagreement. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-01-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-01-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-01-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-01-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-01-TST-005 - Critical invariant. No protected effect occurs without current identity, authority, policy, evidence, verification, and exact-change checks. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-01-TST-006 - Failure containment. Inject root-state uncertainty, Owner identity conflict, replayed Authority or Permission decision, audit-anchor mismatch, replica disagreement. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-01-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-01-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-01-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-01-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.